

MB 511: Python Programming

Course Code	MB 511
Course Name	Python Programming
L-T-P-C	2-0-0-2 (L-T-P-C)
Prerequisites	Preferably having sound knowledge in programming
Intended for	MBA
Distribution	Compulsory
Approval	50th BoA (Jul 2023)

Course Contents

- **Basics of Programming:** This module presents a primer on the building blocks of a program and how to logically sequence the components to perform a complex task. The following topics will be covered: (5 hours)
 - Introduction to Programming
 - Variables, Statements and Conditional Execution
 - Functions
 - Iterations
- **Data Structures:** This module introduces the fundamental data structures in Python and Panda. This module helps the students to learn “How should data be stored in a particular business setting and what are the trade-offs involved?”. The following topics will be covered: (8 hours)
 - Strings and Files
 - Lists and Dictionaries
 - Pandas DataFrame Basics
 - Pandas Data Structure
- **Basic Analysis:** This module introduces the basic techniques in Pandas for plotting, assembling, and handling missing data. The following topics will be covered: (8 hours)
 - Introduction to Plotting
 - Data Assembly
 - Missing Data
- **Data Munging:** Data munging, also known as data wrangling, is the process of transforming raw data into another format with the intent of making it more appropriate for analysis. It is one of the very important steps in data analysis. The following topics will be covered: (7 hours)
 - Tidy Data and Data Types
 - Text Data
 - Pandas Apply and Group-by Operations
- **Lab Exercises** (If applicable): Lab to be conducted on a 2-hour slot. It will be conducted in tandem with the theory course so the topics for problems given in the lab are already initiated in the theory class. The topics taught in the theory course should appropriately be sequenced for synchronization with the laboratory.

Text Books

1. Charles R. Severance, **Python for Everybody: Exploring Data in Python 3**, Amazon Digital Services, 2016.
2. Daniel Y. Chen, **Pandas for Everyone: Python Data Analysis**, Pearson Education, 2018.

References

1. Michael Dawson, **Python Programming for the Absolute Beginners**, Cengage, 2020.